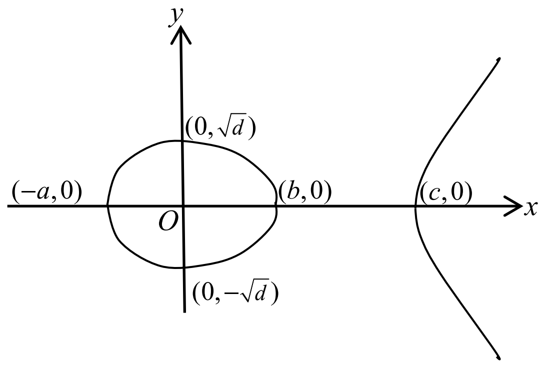
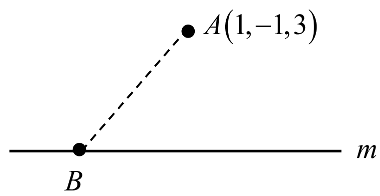
No.	Solutions	Comments
1i)	$\mathrm{f}^2(x)=\mathrm{f}\left(\frac{1}{1-x}\right)=\frac{1}{1-\frac{1}{1-x}}=\frac{1-x}{-x}=1-\frac{1}{x},\ x\neq 1, x\neq 0$ $y=\frac{1}{1-x}\Rightarrow x=1-\frac{1}{y}, x\neq 1, x\neq 0$ $\therefore \mathrm{f}^{-1}(x)=1-\frac{1}{x}$ $\therefore \mathrm{f}^2(x)=\mathrm{f}^{-1}(x), x\neq 1, x\neq 0$	Note that this is a show question hence there is a need to find both $\mathrm{f}^2(x)$ and $\mathrm{f}^{-1}(x)$. Note that $\mathrm{f}^2(x)=\mathrm{f}\left[\mathrm{f}(x)\right]$
1ii)	$\mathrm{f}^2(x)=\mathrm{f}^{-1}(x)$ $\therefore \mathrm{f}^3(x)=\mathrm{ff}^{-1}(x)=x$	Fastest way is to apply answer in (i).
2	$x^2y+xy^2+54=0$ $x^2\frac{\mathrm{d}y}{\mathrm{d}x}+2xy+x\left(2y\frac{\mathrm{d}y}{\mathrm{d}x}\right)+y^2=0$ $\frac{\mathrm{d}y}{\mathrm{d}x}=-1\Rightarrow -x^2+2xy-2xy+y^2=0\Rightarrow y=\pm x$ $y=x\Rightarrow x^3+x^3=-54\Rightarrow x=-3$ $y=-x\Rightarrow -x^3+x^3+54=0$ (no soln) $\therefore(-3,-3)$ is the only point on C at which the gradient is -1.	There is no need to spend time expressing $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x and y before substituting $\frac{\mathrm{d}y}{\mathrm{d}x}=-1$. Note that $x^2=y^2\Rightarrow y=\pm x$ (2 possible solutions!)
3i)	$\underset{\sim}{a}\times\underset{\sim}{b}=\underset{\sim}{0}$ $\Rightarrow$ $\underset{\sim}{a}$ or $\underset{\sim}{b}$ are zero vector or vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are parallel i.e. $\underset{\sim}{a}=\lambda\underset{\sim}{b},\ \lambda\in\mathbb{R}$	Many left out the answer "$\underset{\sim}{a}$ or $\underset{\sim}{b}$ are zero vector"
3ii)	$\underset{\sim}{n}\times\begin{pmatrix}1\\2\\-2\end{pmatrix}=\underset{\sim}{0}\Rightarrow\underset{\sim}{n}=\lambda\begin{pmatrix}1\\2\\-2\end{pmatrix}$ $\Rightarrow$ unit vector $\underset{\sim}{n}=\frac{1}{3}\begin{pmatrix}1\\2\\-2\end{pmatrix}$	The clue in (i) is often ignored.
3iii)	$\frac{\begin{pmatrix}1\\2\\-2\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{3}=-\frac{2}{3}$ $\Rightarrow$ cosine of the acute angle $=\frac{2}{3}$	Question says "… cosine of the acute angle…" $\Rightarrow$ (1) No need to find the angle & (2) $\cos\theta=-\mathrm{ve}$ means that θ is obtuse.

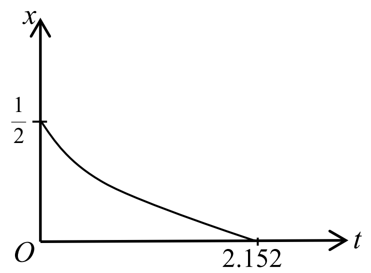
4i)	$y^2 = \mathrm{f}(x)$ 	Not in H2 Math 9758 syllabus (1) Need to label the key points in terms of <u>*a*, *b*, *c* and *d*</u> AND in <u>coordinates</u>. (2) Take note of shape of curve at the *x*-intercepts, tangents to the curve at those points are vertical lines.
4ii)	The tangents to the curve $y^2 = \mathrm{f}(x)$ at the points where it crosses the *x*- axis are parallel to the *y*-axis.	Not in H2 Math 9758 syllabus Question says "… said about the tangents…" $\Rightarrow$ not enough to say gradient is undefined!
5i)	$z = 1+2\mathrm{i}$ $\Rightarrow z^2 = 1+4\mathrm{i}+4\mathrm{i}^2 = -3+4\mathrm{i}$ $\frac{1}{z^3} = \frac{1}{(1+2\mathrm{i})(-3+4\mathrm{i})}$ $= \frac{1}{-3-2\mathrm{i}-8}$ $= \left(\frac{1}{-11-2\mathrm{i}}\right)\left(\frac{-11+2\mathrm{i}}{-11+2\mathrm{i}}\right)$ $= \frac{-11+2\mathrm{i}}{11^2+2^2}$ $= -\frac{11}{125}+\frac{2}{125}\mathrm{i}$	Question says "Without using a calculator…" hence should not be using polar form.
5ii)	$pz^2 + \frac{q}{z^3} = p(-3+4\mathrm{i}) + q\left(-\frac{11}{125}+\frac{2}{125}\mathrm{i}\right)$ $= \left(-3p-\frac{11}{125}q\right)+\left(4p+\frac{2}{125}q\right)\mathrm{i}$ $pz^2+\frac{q}{z^3}$ is real $\Rightarrow 4p+\frac{2}{125}q = 0 \quad \therefore q = -250p$ $pz^2+\frac{q}{z^3} = -3p-\frac{11}{125}(-250p) = 19p$	
6ai)	Let Q_n be the statement $p_n = \frac{1}{3}(7-4^n),\ n \in \mathbb{Z}^+$ LHS of $\mathrm{Q}_1 = p_1 = 1$ (given) RHS of $\mathrm{Q}_1 = \frac{1}{3}(7-4^1) = 1$ $\therefore \mathrm{Q}_1$ is true. Assume Q_k is true for some $k \in \mathbb{Z}^+$ i.e. $p_k = \frac{1}{3}(7-4^k)$	Not in H2 Math 9758 syllabus

	To show that Q_{k+1} is true i.e. $p_{k+1}=\frac{1}{3}\left(7-4^{k+1}\right)$ LHS of $Q_{k+1}=p_{k+1}$ $=4p_k-7$ $=4\left[\frac{1}{3}\left(7-4^k\right)\right]-7$ $=\frac{1}{3}\left(28-4^{k+1}-21\right)$ $=\frac{1}{3}\left(7-4^{k+1}\right)=\text{RHS of } p_{k+1}$ $\therefore Q_k$ is true $\Rightarrow Q_{k+1}$ is true Since Q_1 is true and Q_k is true $\Rightarrow Q_{k+1}$ is true. $\therefore$ by mathematical induction, Q_n is true for all positive integers.	
6aii)	$\sum_{r=1}^{n} p_r=\sum_{r=1}^{n}\frac{1}{3}\left(7-4^r\right)$ $=\frac{7}{3}n-\frac{1}{3}\sum_{r=1}^{n}4^r$ $=\frac{7}{3}n-\frac{1}{3}\cdot\frac{4\left(1-4^n\right)}{\left(1-4\right)}$ $=\frac{7}{3}n+\frac{4}{9}\left(1-4^n\right)$	
6bi)	$S_n=1-\frac{1}{(n+1)!}$ As $n\to\infty$, $(n+1)!\to\infty\Rightarrow\frac{1}{(n+1)!}\to 0$ $\therefore S_n\to 1$, a finite value as $n\to\infty$. $\therefore\sum u_r$ converges and the value of the sum to infinity $=1$	
6bii)	$u_n=S_n-S_{n-1}$ $=1-\frac{1}{(n+1)!}-\left[1-\frac{1}{n!}\right]$ $=\frac{1}{n!}-\frac{1}{(n+1)!}$ $=\frac{n+1-1}{(n+1)!}=\frac{n}{(n+1)!}$	Question says "… in simplified form." $\Rightarrow$ there is a need to simplify the expression to a single fraction.
7i)	$\mathrm{f}(\alpha)=0\Rightarrow\alpha=\pm 1.885245$ (from GC) $\alpha\approx 1.885$ (to 3 d.p.) $(\because\alpha>0)$ $\mathrm{f}(\beta)=-7\Rightarrow\beta^6-3\beta^4=0$ $\Rightarrow\beta^4\left(\beta^2-3\right)=0\Rightarrow\beta=0$ or $\beta=\pm\sqrt{3}$ $\Rightarrow\ \beta=\sqrt{3}$ since $\beta>0$	
7ii)	$\int_{\sqrt{3}}^{1.885}\left(x^6-3x^4-7\right)\,\mathrm{d}x$ $=-0.597$ (to 3 d.p.)	This is a straigtforward integral unrelated to area.

7iii)	Area of the required finite region $=\int_0^{\sqrt{3}}\left[-7-\left(x^6-3x^4-7\right)\right]\mathrm{d}x$ $=\left[-\frac{x^7}{7}+\frac{3x^5}{5}\right]_0^{\sqrt{3}}$ $=27\sqrt{3}\left(-\frac{1}{7}+\frac{1}{5}\right)=\frac{54\sqrt{3}}{35}$	Note that region is bounded between 2 graphs: $y_{\text{upper}}=-7$ and $y_{\text{lower}}=x^6-3x^4-7$ $\Rightarrow$ area $=\int_0^{\sqrt{3}}\left[y_{\text{upper}}-y_{\text{lower}}\right]\mathrm{d}x$ Terms such as $\left(\sqrt{3}\right)^5$ must be simplified in answers.
7iv)	$\mathrm{f}(-x)=(-x)^6-3(-x)^4-7$ $=x^6-3x^4-7$ $=\mathrm{f}(x)$ Since the coefficients of $\mathrm{f}(x)$ are all real, so $\mathrm{f}(x)=0$ has 2 real roots $(\alpha$ and $-\alpha)$ and 2 pairs of conjugate roots. Let $a\pm\mathrm{i}b$ be 1 pair of conjugate roots. Since $\mathrm{f}(x)=\mathrm{f}(-x)$, $\therefore -\left(a\pm\mathrm{i}b\right)$ will be the other pair of conjugate roots. $\therefore$ the non-real roots are $a+\mathrm{i}b$ $a+\mathrm{i}b$, $a-\mathrm{i}b$, $-a-\mathrm{i}b$ and $-a+\mathrm{i}b$	The graph of $y=\mathrm{f}(x)$ where $x\geq 0$, shows 1 real root, α. Hence from the result, $\mathrm{f}(x)=\mathrm{f}(-x)$, we can deduce the 2nd real root $-\alpha$.
8i)	$\int\frac{1}{\sqrt{9-x^2}}\,\mathrm{d}x=\sin^{-1}\left(\frac{x}{3}\right)+C$	Remember to include the arbitrary constant.
8ii)	$\frac{1}{\sqrt{9-x^2}}=\left(9-x^2\right)^{-\frac{1}{2}}$ $=\frac{1}{3}\left[1-\frac{x^2}{9}\right]^{-\frac{1}{2}}$ $=\frac{1}{3}\left[1+\left(-\frac{1}{2}\right)\left(-\frac{x^2}{9}\right)+\frac{-\frac{1}{2}\left(-\frac{3}{2}\right)}{2!}\left(-\frac{x^2}{9}\right)^2+\frac{-\frac{1}{2}\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}\left(-\frac{x^2}{9}\right)^3\right]+\ldots$ $=\frac{1}{3}\left(1+\frac{x^2}{18}+\frac{3x^4}{648}+\frac{5x^6}{11664}+\ldots\right)$ $=\frac{1}{3}+\frac{1}{54}x^2+\frac{1}{648}x^4+\frac{5}{34992}x^6+\ldots$	Question says "… binomial expansion…" $\Rightarrow$ should use the standard series in MF26 instead of repeated differentation which often leads to wrong answer.
8iii)	$\sin^{-1}\left(\frac{x}{3}\right)+C=\int\left(\frac{1}{3}+\frac{1}{54}x^2+\frac{1}{648}x^4+\frac{5}{34992}x^6+\ldots\right)\mathrm{d}x$ $\therefore\sin^{-1}\left(\frac{x}{3}\right)+C=\frac{1}{3}x+\frac{1}{54}\cdot\frac{x^3}{3}+\frac{1}{648}\cdot\frac{x^5}{5}+\frac{5}{34992}\cdot\frac{x^7}{7}+\ldots$ When $x=0$, $\sin^{-1}(0)+C=0\Rightarrow C=0$ $\therefore$ Maclaurin series for $\sin^{-1}\left(\frac{x}{3}\right)$ $=\frac{1}{3}x+\frac{1}{162}x^3+\frac{1}{3240}x^5+\frac{5}{244944}x^7+\ldots$	Do not omit the arbitrary constant. There is a need to find the value of this constant.

9i)	Planes p and q are perpendicular $\Rightarrow \begin{pmatrix}1\\2\\-3\end{pmatrix}$ is parallel to plane q. Plane q contains the line l $\Rightarrow \begin{pmatrix}2\\-1\\4\end{pmatrix}$ is parallel to plane q. $\begin{pmatrix}1\\2\\-3\end{pmatrix}\times\begin{pmatrix}2\\-1\\4\end{pmatrix}=\begin{pmatrix}5\\-10\\-5\end{pmatrix}=5\begin{pmatrix}1\\-2\\-1\end{pmatrix}$ $\Rightarrow \begin{pmatrix}1\\-2\\-1\end{pmatrix}$ is a vector normal to plane q. Equation of plane q is $\mathbf{r}\cdot\begin{pmatrix}1\\-2\\-1\end{pmatrix}=\begin{pmatrix}1\\-1\\3\end{pmatrix}\cdot\begin{pmatrix}1\\-2\\-1\end{pmatrix}$ i.e. $\mathbf{r}\cdot\begin{pmatrix}1\\-2\\-1\end{pmatrix}=0$ Cartesian equation of q is $x-2y-z=0$	Note that $\mathbf{r}=\mathbf{a}+\alpha\mathbf{b}+\beta\mathbf{c}$ is not cartesian form.
9ii)	Plane p: $x+2y-3z=12 -----(1)$ Plane q: $x-2y-z=0------(2)$ Solving using GC: $x=6+2t,\ y=3+\frac{1}{2}t,\ z=t$ $\therefore$ a vector equation of the line m is $\mathbf{r}=\begin{pmatrix}6\\3\\0\end{pmatrix}+\lambda\begin{pmatrix}4\\1\\2\end{pmatrix},\ \lambda\in\mathbb{R}$	
9iii)	$\overrightarrow{OB}=\begin{pmatrix}6+4\lambda\\3+\lambda\\2\lambda\end{pmatrix}$ and $\overrightarrow{OA}=\begin{pmatrix}1\\-1\\3\end{pmatrix}$ Let $S^2=AB^2$ $S^2=\left\|\overrightarrow{AB}\right\|^2$ $=(5+4\lambda)^2+(4+\lambda)^2+(2\lambda-3)^2$ $=25+40\lambda+16\lambda^2+16+8\lambda+\lambda^2+4\lambda^2-12\lambda+9$ $=50+36\lambda+21\lambda^2$ $2S\frac{\mathrm{d}S}{\mathrm{d}\lambda}=36+42\lambda\Rightarrow S\frac{\mathrm{d}S}{\mathrm{d}\lambda}=18+21\lambda$ $\frac{\mathrm{d}S}{\mathrm{d}\lambda}=0\Rightarrow\lambda=-\frac{6}{7}$	 Note that the point on m which is nearest to A $\Rightarrow AB$ is the shortest.

	$S\frac{\mathrm{d}^2S}{\mathrm{d}\lambda^2}+\left(\frac{\mathrm{d}S}{\mathrm{d}\lambda}\right)^2=21$ Since $S>0$ and $\frac{\mathrm{d}S}{\mathrm{d}\lambda}=0$ when $\lambda=-\frac{6}{7}$, $\therefore \frac{\mathrm{d}^2S}{\mathrm{d}\lambda^2}>0$ $\Rightarrow S=AB$ is minimum when $\lambda=-\frac{6}{7}$ Coordinates of the point on m which is nearest to A is $\left(\frac{18}{7},\frac{15}{7},-\frac{12}{7}\right)$.	
10i)	$\frac{\mathrm{d}x}{\mathrm{d}t}=k\left(1+x-x^2\right),\ 0\le x\le\frac{1}{2}$ Given $x=\frac{1}{2}$, $\frac{\mathrm{d}x}{\mathrm{d}t}=-\frac{1}{4}$ $\therefore -\frac{1}{4}=k\left(1+\frac{1}{2}-\frac{1}{4}\right)$ $\Rightarrow \frac{5}{4}k=-\frac{1}{4}\Rightarrow k=-\frac{1}{5}$	This is a show question hence you should avoid missing out steps in such a situation.
10ii)	$1+x-x^2=\frac{5}{4}-\left(x-\frac{1}{2}\right)^2$ $\therefore \frac{\mathrm{d}x}{\mathrm{d}t}=-\frac{1}{5}\left[\frac{5}{4}-\left(x-\frac{1}{2}\right)^2\right]$ $\Rightarrow \int -\frac{1}{5}\ \mathrm{d}t=\int \frac{1}{\left(\frac{\sqrt{5}}{2}\right)^2-\left(x-\frac{1}{2}\right)^2}\ \mathrm{d}x$ $-\frac{1}{5}t=\frac{1}{2\left(\frac{\sqrt{5}}{2}\right)}\ln\left\|\frac{\frac{\sqrt{5}}{2}+\left(x-\frac{1}{2}\right)}{\frac{\sqrt{5}}{2}-\left(x-\frac{1}{2}\right)}\right\|+C$ $t=-\sqrt{5}\ln\left\|\frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x}\right\|+C$ Since $0\le x\le\frac{1}{2}$, $\frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x}>0$ $t=-\sqrt{5}\ln\left(\frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x}\right)+C$ When $t=0$, $x=\frac{1}{2}$, $\therefore 0=\frac{1}{\sqrt{5}}\ln 1+C\Rightarrow C=0$ $t=-\sqrt{5}\ln\left(\frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x}\right)$	Remember to remove the modulus with reason given.
10iiia)	When $x=\frac{1}{4}$, $t=-\sqrt{5}\ln\left(\frac{\sqrt{5}-1+\frac{1}{2}}{\sqrt{5}+1-\frac{1}{2}}\right)=\sqrt{5}\ln\left(\frac{2\sqrt{5}+1}{2\sqrt{5}-1}\right)$	
10iiib)	When $x=0$, $t=-\sqrt{5}\ln\left(\frac{\sqrt{5}-1}{\sqrt{5}+1}\right)\approx 2.15204\approx 2.152$ (to 3 d.p.)	

10iv)	$t=-\sqrt{5}\ln\left(\frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x}\right)$ $\Rightarrow\left(\frac{\sqrt{5}-1+2x}{\sqrt{5}+1-2x}\right)=\mathrm{e}^{-\frac{1}{\sqrt{5}}t}$ $\sqrt{5}-1+2x=\left(\sqrt{5}+1\right)\mathrm{e}^{-\frac{1}{\sqrt{5}}t}-2x\mathrm{e}^{-\frac{1}{\sqrt{5}}t}$ $2x\left(1+\mathrm{e}^{-\frac{1}{\sqrt{5}}t}\right)=\left(\sqrt{5}+1\right)\mathrm{e}^{-\frac{1}{\sqrt{5}}t}-\left(\sqrt{5}-1\right)$ $x=\frac{\left(\sqrt{5}+1\right)\mathrm{e}^{-\frac{1}{\sqrt{5}}t}-\left(\sqrt{5}-1\right)}{2\left(1+\mathrm{e}^{-\frac{1}{\sqrt{5}}t}\right)}$ 	
11i)	$V=\frac{1}{2}\left(\frac{4}{3}\pi r^3\right)+\frac{1}{3}\pi r^2 h$ $\therefore V=\frac{2}{3}\pi r^3+\frac{1}{3}\pi r^2\left(16-r^2\right)^{\frac{1}{2}}\quad \because h^2+r^2=16$ $\frac{\mathrm{d}V}{\mathrm{d}r}=2\pi r^2+\frac{2}{3}\pi r\left(16-r^2\right)^{\frac{1}{2}}+\frac{1}{3}\pi r^2\cdot\frac{1}{2}\left(16-r^2\right)^{-\frac{1}{2}}(-2r)$ $=2\pi r^2+\frac{2}{3}\pi r\left(16-r^2\right)^{\frac{1}{2}}-\frac{1}{3}\pi r^3\left(16-r^2\right)^{-\frac{1}{2}}$ V is maximum $\Rightarrow\frac{\mathrm{d}V}{\mathrm{d}r}=0$ $\Rightarrow 2\pi r^2+\frac{2}{3}\pi r\left(16-r^2\right)^{\frac{1}{2}}-\frac{1}{3}\pi r^3\left(16-r^2\right)^{-\frac{1}{2}}=0$ $\Rightarrow 6\pi r^2\left(16-r^2\right)^{\frac{1}{2}}+2\pi r\left(16-r^2\right)-\pi r^3=0$ Since $\pi r\neq 0$, $\therefore 6r\left(16-r^2\right)^{\frac{1}{2}}=3r^2-32$ ---- (1) Squaring (1): $36r^2\left(16-r^2\right)=9r^4-192r^2+1024$ $\Rightarrow 45r^4-768r^2+1024=0$ Since $r=r_1$ gives maximum value of V, $\therefore r_1$ satisfies the equation $45r^4-768r^2+1024=0$	Tip: If intended to square equation to get rid of square root, algebraically easier if $\left(16-r^2\right)^{\frac{1}{2}}$ is made the subject before squaring both sides.
11ii)	From GC: $r=\pm 3.950797,\ \pm 1.20742$ Since $r > 0$, $r = 3.951, 1.207$ (to 3 d.p.)	
11iii)	When $r = 1.20742$, $\frac{\mathrm{d}V}{\mathrm{d}r}=18.320\neq 0\Rightarrow r=1.207$ does not give	Question says "…does not give a stationary value…" $\Rightarrow$ need to check whether

	a stationary value of V. $\therefore r_1 = 3.951$ $h=\sqrt{16-3.950797^2} \approx 0.625$	$\frac{\mathrm{d}V}{\mathrm{d}r}=0$. Important that values of r is NOT substituted back into $45r^4-768r^2+1024=0$!
11iv)	$V=\frac{2}{3}\pi r^3+\frac{1}{3}\pi r^2\left(16-r^2\right)^{\frac{1}{2}},\ 0<r<4$